NASIR MAHMOOD ABBASI, PhD

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RESEARCH PROFILE

Bioinformatics Engineer & Computational Biologist (PhD, Cancer Biology) bridging computational pipeline engineering with experimental oncology to decipher tumor heterogeneity and identify targetable disease vulnerabilities. Brings 9+ years managing clinical multi-omic datasets, administering Linux/Slurm HPC infrastructure, and deploying automated workflows (Python, R, Snakemake, Nextflow, Docker) across single-cell multi-omics, transcriptomics, and cancer genomics to deliver actionable biological insights.

- **Single-Cell Multi-Omics:** Autonomously processes and integrates scRNA-seq, surface proteomics (CITE-seq), and TCR clonality (Seurat, Scanpy, Harmony, inferCNV) to resolve clonal evolution and tumor heterogeneity across patient cohorts.
- **Developmental Trajectory Modeling:** Infers continuous lineage differentiation hierarchies and pseudotime state transitions using Monocle3, Slingshot, and PAGA against reference developmental maps.
- **Systems Leadership & HPC Infrastructure:** Administered dedicated Linux Slurm HPC cluster at IRBI, managing multi-user resource scheduling, storage allocation, and containerized deployment (Docker, Singularity, Conda).
- **Cross-Functional Collaboration & Validation:** Partners closely with wet-lab experimentalists and clinicians to experimentally validate computational findings; reconstructed regulatory networks (SCENIC) and metabolic scores (AUCell), predicting a candidate therapeutic target validated through in vitro pharmacological inhibition.
- **Open-Source Tools & Platforms:** Authors and maintains public reproducible HPC workflows on GitHub, including OmicsFlow-BulkRNAseq (Nextflow, Snakemake) and Snakemake-PacBio-Assembly for automated long-read genome assembly.

EDUCATION

PhD in Cancer Biology, Université de Bordeaux *2023 - 2026*
France
Research Unit: INSERM U1312, Bordeaux Research in Translational Oncology (BRIC)

- *Supervisors:* Prof. Audrey Gros (Director) and Dr. Slim Karkar (Co-Director)
- *Defense Status:* Defended 23 June 2026 | Manuscript online August 2026
- *Thesis Title: "**Deciphering Sézary syndrome tumoral heterogeneity through a bioinformatic approach**"*

Masters in Bioinformatics, Uppsala University *2010 - 2012*
Sweden

BS (Hons) in Bioinformatics, COMSATS University *2005 - 2009*
Pakistan

TECHNICAL & COMPUTATIONAL SKILLS

Single-cell analysis: Seurat, Scanpy, SingleCellExperiment, SCTransform, Harmony, DoubletFinder, Scrublet, Azimuth, SingleR, scType, Monocle3, Slingshot, PAGA, scVelo, Velocyto

Genomics & Spatial: WES, WGS, Samtools, Bedtools, bulk RNA-seq, metatranscriptomics, PacBio HiFi, Nanopore, SPAdes, Kraken2, Salmon, eggNOG-mapper, BioNano, GATK, inferCNV | Spatial: 10x Visium, NanoString CosMx, Squidpy, Seurat Spatial

Pipelines, Containers & HPC: Nextflow, Snakemake, Slurm HPC cluster administration (32-node, 1,024-core), Docker, Singularity, Conda, Git / GitHub Actions CI/CD, Linux/Unix, Galaxy administration

Languages & Scripting: R (Bioconductor, Tidyverse, ComplexHeatmap), Python, Bash shell scripting, SQL, SPARQL, Perl, C, JavaScript, PHP

Metabolic Profiling & Networks: scMetabolism, AUCell, SCENIC, GENIE3, DoRothEA, LIANA, CellChat, UniProt, MetaCyc, ChEBI, Rhea

Data Visualization & Reporting: ggplot2, ComplexHeatmap, SCpubr, Nebulosa, dittoSeq, Quarto / RMarkdown automated reporting, Plotly

Statistics & Quality Control: DESeq2, edgeR, MAST, LIBRA (pseudobulk), Wilcoxon rank-sum test, FastQC, MultiQC

RESEARCH EXPERIENCE

PhD Researcher, Université de Bordeaux / INSERM U1312 (BRIC) ***2023 - 2026***

Supervisors: *Prof. Audrey Gros, Dr. Slim Karkar*

- Engineered end-to-end multi-omic pipelines integrating scRNA-seq, surface proteomics (CITE-seq), and TCR clonality across 7 patient-derived cell models in Sézary syndrome.
- Integrated WES somatic variant calling, BioNano optical genome mapping, and single-cell transcriptomes on cluster Slurm HPC, mapping structural rearrangements and clonal dynamics.
- Uncovered transcription factor networks (SCENIC) and metabolic programs; predicted candidate therapeutic target validated through in vitro pharmacological assays (Biomarker Research, under review).

Research Engineer, IRBI Insect Biology Research Institute ***2022 - 2023***

Tours, France

- Administered dedicated 32-node (1,024-core) Linux Slurm HPC cluster; deployed automated PacBio HiFi and Oxford Nanopore assembly workflows across 20 sequencing libraries using Docker and Singularity.

Research Engineer, INRAE ***2019 - 2021***

Clermont-Ferrand, France

- Benchmarked small RNA-seq analytical pipelines (siRNA/miRNA) across plant model genomes (Brachypodium, maize, wheat); administered enterprise Galaxy platform supporting multi-user transcriptomics and metatranscriptomics.

Research Assistant, Hong Kong University of Science and Technology (HKUST) ***2017 - 2018***

Hong Kong, China

- Benchmarked GATK, somatic SNV, and indel calling workflows across patient exomes and whole genomes, establishing sensitivity and specificity baselines on HPC clusters.

Earlier research roles ***2009 - 2017***

Universidad Miguel Hernández de Elche | Evolutionary Biology Centre | SciLifeLab | COMSATS University

- Marine prokaryotic genome analysis and web development | **Universidad Miguel Hernández de Elche**
- Behavioral genetics and epigenetics research (BMC Evolutionary Biology. 2017) | **Evolutionary Biology Centre, Sweden**
- Bacterial genomics, contributing to identification of a new Capnocytophaga species (Scientific Reports. 2016) | **Karolinska Hospital, Sweden**
- Exome sequencing for rare disease variant discovery and involvement in Wetlab experiments (Molecular Vision. 2013) | **COMSATS University**

TEACHING

Teaching Assistant (TP - Travaux Pratiques), Université de Bordeaux ***2023 - 2025***

- Delivered practical/laboratory teaching sessions (TP) each year of the PhD: 20 HETD equivalent hours in 2023–2024, 10 HETD equivalent hours in 2024–2025

Instructor of Bioinformatics, Virtual University of Pakistan ***2017***

- Taught bioinformatics coursework and supervised laboratory exercises in molecular biology, microbiology, and cell biology

PUBLICATIONS

Under Review

Boccelari L, Gravelle P, Tosolini M, **Abbasi N**, Le Morvan V, et al. Primary Cutaneous Diffuse Large B Cell Lymphoma, Leg Type, Induces Dynamic Transcriptional Reprogramming of Immunosuppressive Macrophages. *Cancer Immunology Research.* **Under review**

Abbasi NM, *Paz Del Socorro T, Gravelle P, Tosolini M, Prochazkova-Carlotti M, Pham Ledard A, Beylot-Barry M, Isnard A, Roussel A, Boccelari L, Guéry MA, Merlio JP, Poglio S, Laurent C, Karkar S, Gros A. Single-cell multi-omics of Sézary cell lines reveals transcriptomic heterogeneity and identifies a novel metabolic vulnerability and therapeutic target. Biomarker Research.* ***Under review***

In Preparation

Cherfan C, Deslande M, Prochazkova-Carlotti M, **Abbasi N**, Sharanek A, Pham-Ledard A, Poglio S, Gros A, Merlio J-P, Bresson-Bepoldin L, Beylot-Barry M, Chevret E. Oxidative Phosphorylation: A Potential Achilles' Heel of Sézary Cells

Published

Ajmal M, Khan M, Neveling K, Tayyab A, Jaffar S, Sadeque A, Ayub H, **Abbasi N**, Riaz M, Micheal S, Gilissen C, Ali S, Azam M, Collin R, Cremers F, Qamar R. Exome sequencing identifies a novel and a recurrent BBS1 mutation in Pakistani families with Bardet-Biedl syndrome. *Molecular Vision.* 2013;19:644-653

Zangenah S, **Abbasi N**, Anders FA, Bergman P. Whole Genome Sequencing Reveals Novel Species of genus Capnocytophaga Isolated From Dog and Cat Bite Wounds in Humans. *Scientific Reports.* 2016

Zajitschek S, Herbert-Read JE, **Abbasi N**, Immler S. Paternal personality and social status influence offspring personality in offsprings. *BMC Evolutionary Biology.* 2017;17:157

LANGUAGES

English: Fluent | **French:** Basic | **Urdu:** Native

CONFERENCES & PRESENTATIONS

- “Single-cell dissection of Sézary syndrome heterogeneity” | Flash Talk & Poster, SBM Day conference, France (September 2024)
- “Tumor heterogeneity and regulatory programs in cutaneous lymphoma” | Poster, SBM Day conference, France (September 2025)
- Poster Presentation |Doctoral School Day, Université de Bordeaux, France (2025)
- gdec-bioinfo-workshop#2 | Tidyverse / R Markdown for Genomics, training workshop, France
- The Hong Kong Epigenomics Project | seminar series, Hong Kong
- The Mathematics of Deep Learning | seminar, Hong Kong

REFERENCES

References will be provided on request.